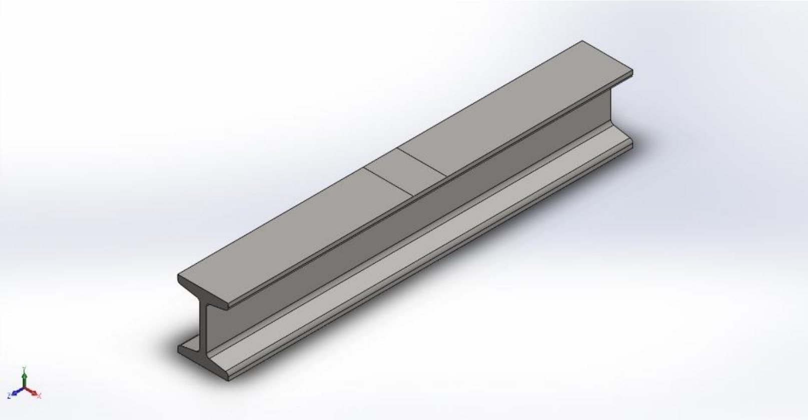
design for intership
Phone: 7982146850



Simulation of beam

Date: 25 June 2026
Designer: Karan Kumar
Study name: I beam
Analysis type: Static

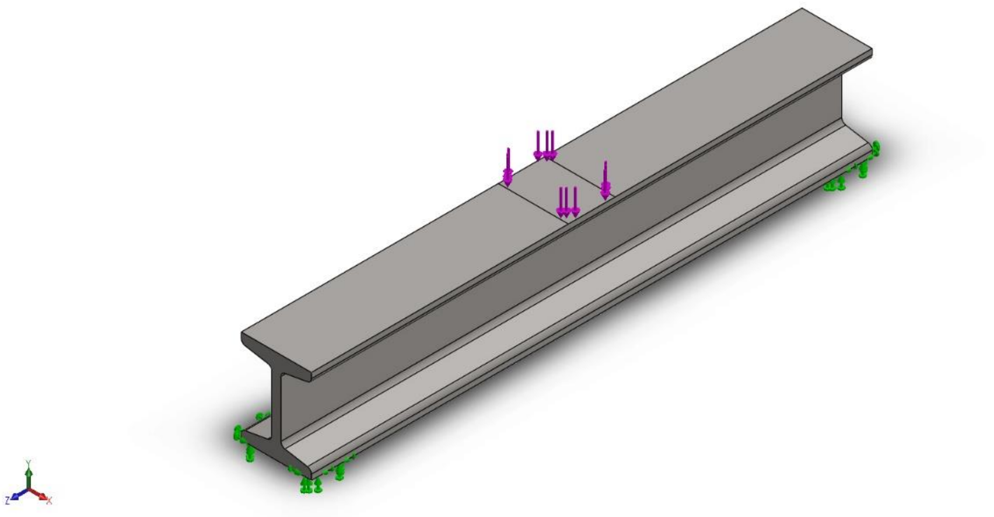
Table of Contents

Description

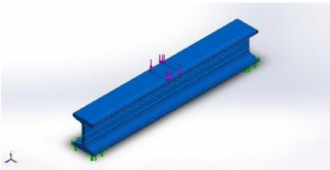
No Data

Assumptions

Model Information



Model name: beam
Current Configuration: Default

Solid Bodies			
Document Name and Reference	**Treated As**	**Volumetric Properties**	**Document Path/Date Modified**
Fillet1 	Solid Body	**Mass:12.828 kg** **Volume:0.00160349 m^3** **Density:8,000 kg/m^3** **Weight:125.714 N**	**C:\Users\Abhay Kumar\OneDrive\Desktop\beam.SLDPRT** Jun 25 16:58:10 2026

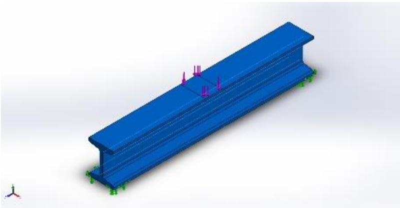
Study Properties

Study name	I beam
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\Abhay Kumar\OneDrive\Desktop)

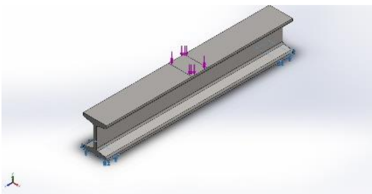
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

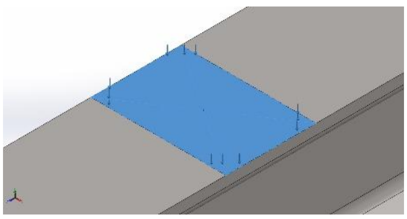
Model Reference	Properties	Components
	**Name:** AISI 304 **Model type:** Linear Elastic Isotropic **Default failure criterion:** Max von Mises Stress **Yield strength:** 2.06807e+08 N/m^2 **Tensile strength:** 5.17017e+08 N/m^2 **Elastic modulus:** 1.9e+11 N/m^2 **Poisson's ratio:** 0.29 **Mass density:** 8,000 kg/m^3 **Shear modulus:** 7.5e+10 N/m^2 **Thermal expansion coefficient:** 1.8e-05 /Kelvin	SolidBody 1(Fillet1)(Part3)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 2 face(s) **Type:** Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	0.405983	9,806.57	-0.47546	9,806.57
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		**Entities:** 1 face(s) **Type:** Apply normal force **Value:** 1,000 kgf

Connector Definitions
No Data

Contact Information
No Data

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	23.4142 mm
Minimum element size	4.68284 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	62442
Total Elements	36799
Maximum Aspect Ratio	4.264
% of elements with Aspect Ratio < 3	99.4
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:00:03
Computer name:	

Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.405983	9,806.57	-0.47546	9,806.57

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	3.71467	2.66988	9.86415	10.8733

Free body moments

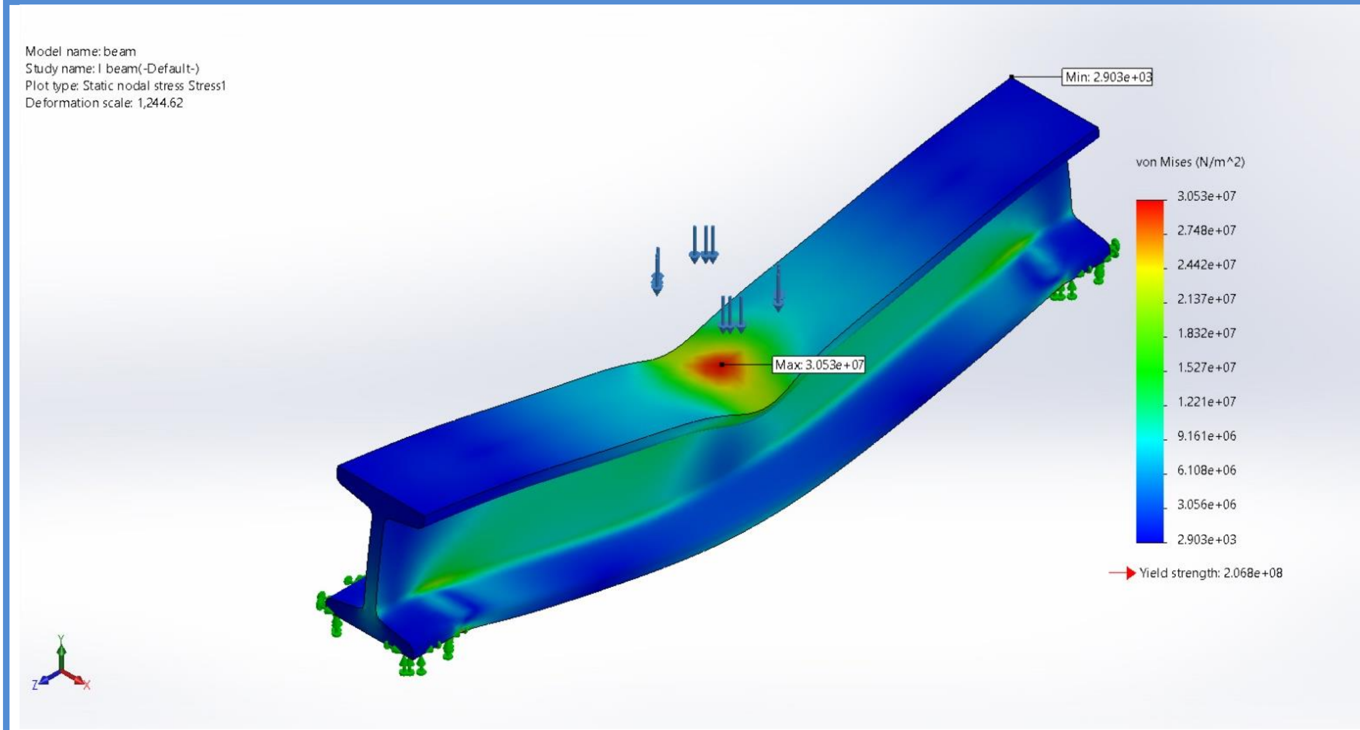
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

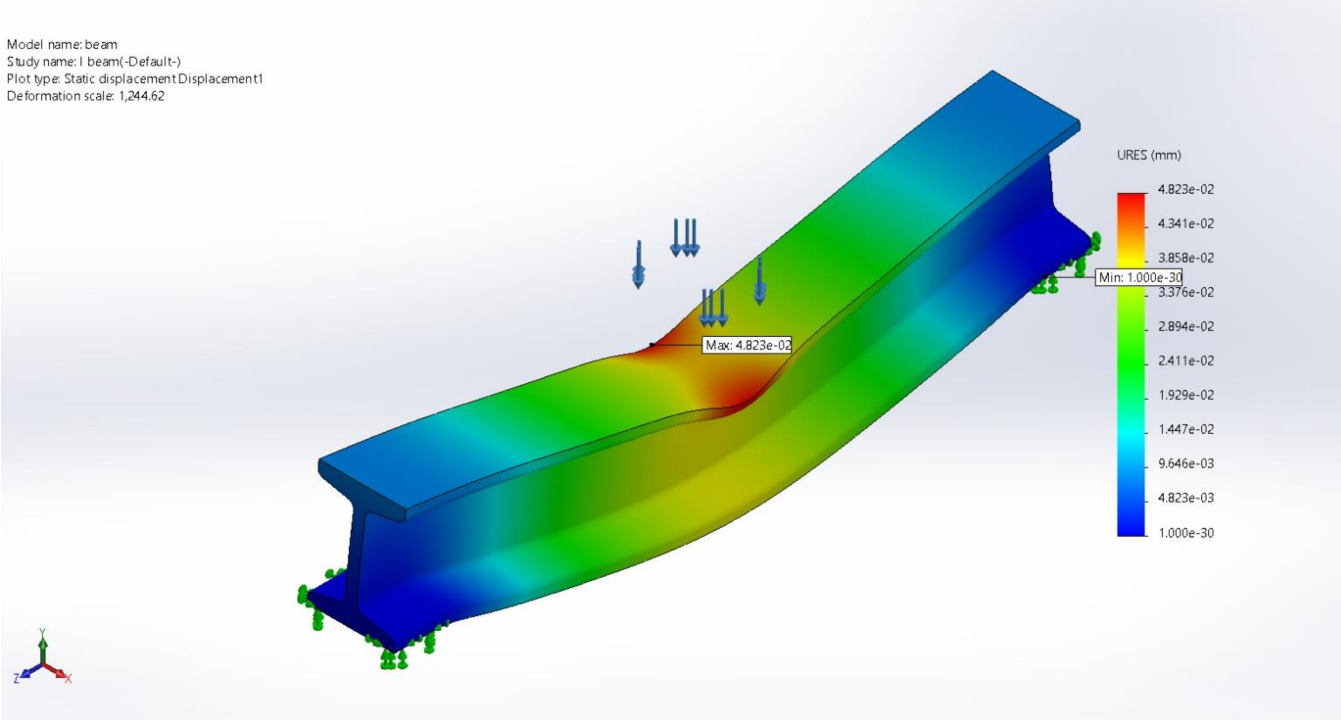
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	2.903e+03N/m^2 Node: 10	3.053e+07N/m^2 Node: 4172



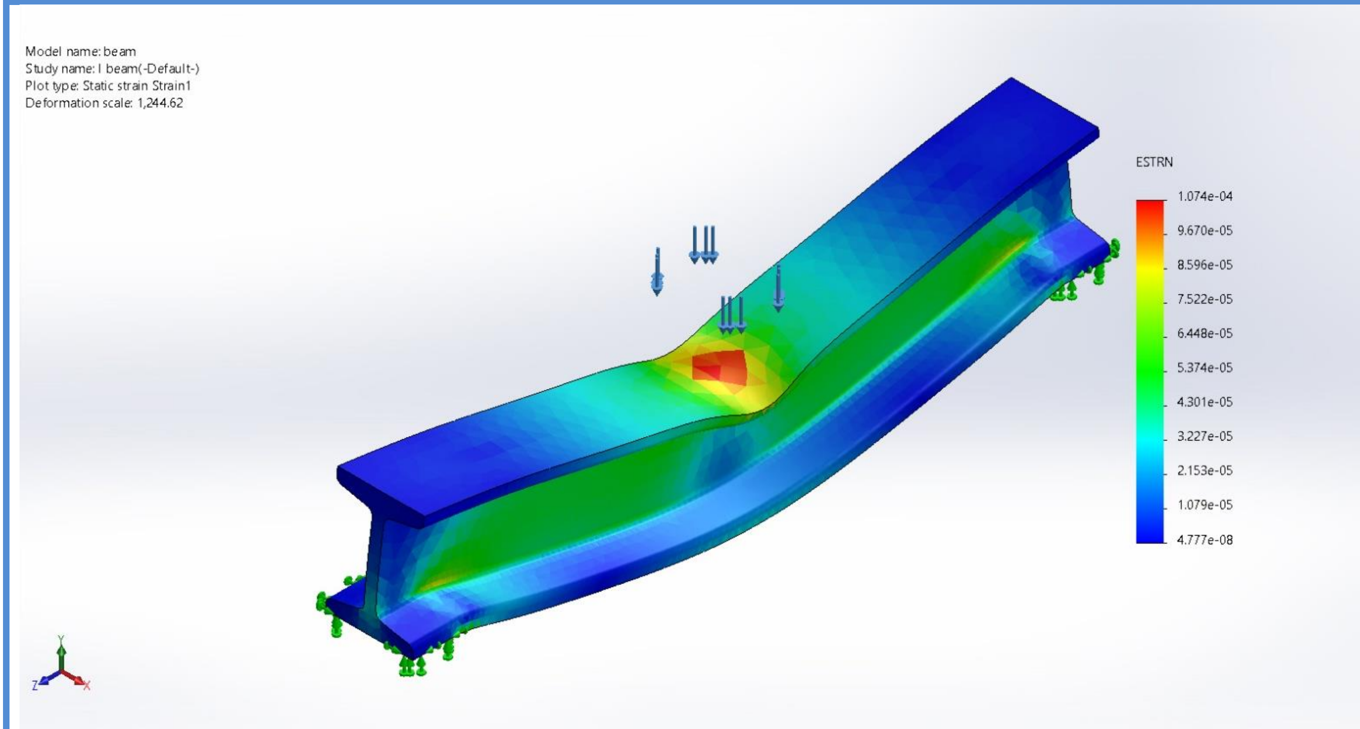
beam-I beam-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 1	4.823e-02mm Node: 62083



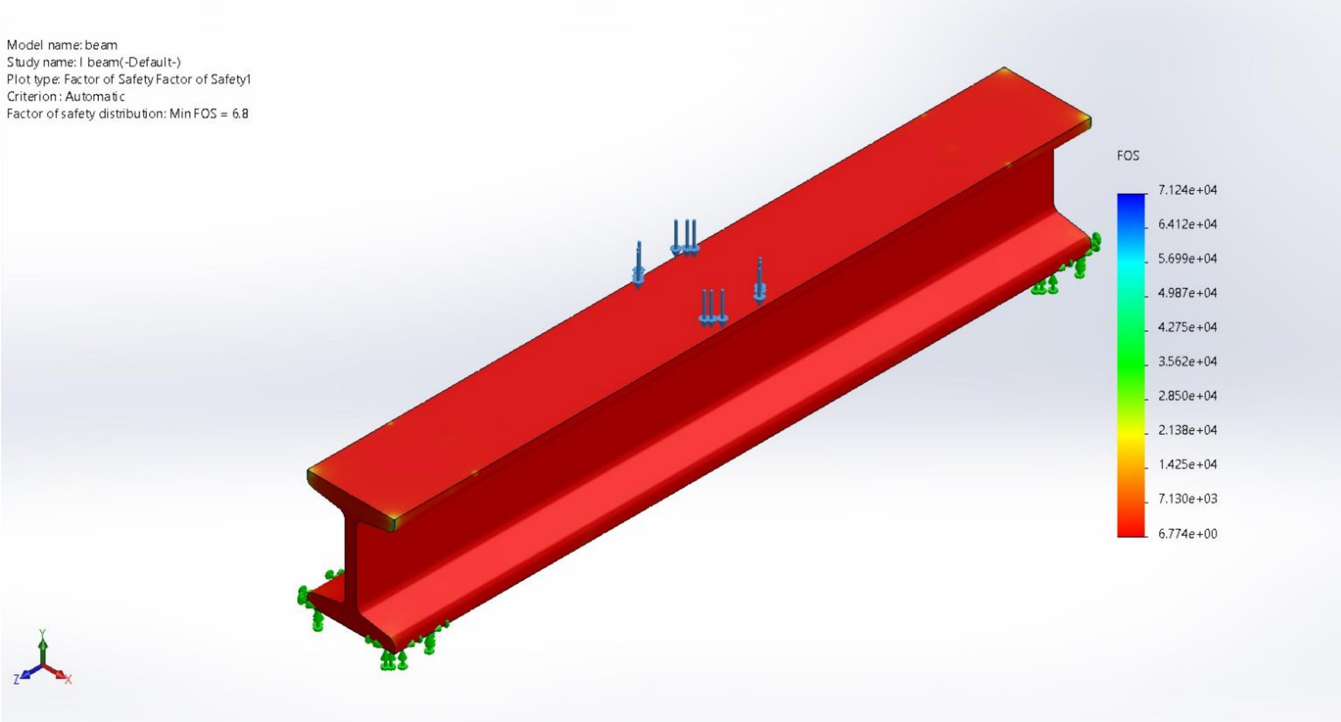
beam-I beam-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	4.777e-08 Element: 17710	1.074e-04 Element: 10417



beam-I beam-Strain-Strain1

Name	Type	Min	Max
Factor of Safety1	Automatic	6.774e+00 Node: 4172	7.124e+04 Node: 10



beam-I beam-Factor of Safety-Factor of Safety1

Conclusion